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# **Global Human Development: A Statistical Exploration of HDI Data (2020-2022)**

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**Submission Date:** 1/10/2026

**GITHUB Link:** [https://github.com/Sujar01/AI\\_Assessment.git](https://github.com/Sujar01/AI_Assessment.git)

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# Introduction

2 The Human Development Index (HDI) is a composite indicator used to compare the overall level of human development across countries by combining health, education, and income into a unique measure of progress. It is maintained by the UNDP and is widely used in global reports to highlight gaps in average life span, education levels, and average income per person between nations. In contrast to purely economic metrics, HDI emphasizes a more broad view of well-being by capturing both social and economic aspects of development.

In this report, the HDI dataset is analyzed using statistical and exploratory data analysis techniques to better understand recent global development patterns. The analysis focuses on four main components: a detailed examination of HDI values for the year 2022, an exploration of HDI trends from 2020 to 2022, a closer inspection of development indicators in South Asian countries, and a relative study between South Asia and the Middle East over the same period. By organizing the work around these themes, the report aims to highlight how levels of human development vary across countries and regions, and how recent years have shaped these trajectories, especially in the context of contemporary global challenges.

## 7 Problem 1A

### Single Year HDI Exploration (2022)

#### 1.1: Data Overview & Cleaning

The 2022 subset contained 206 country observations. Initial inspection revealed missing values in several columns, particularly in economic indicators. Countries with insufficient HDI data were removed to ensure research accuracy. Data type conversions were applied to numeric columns, and duplicate entries were eliminated.

## 1.2: Descriptive Statistics

### Summary Statistics for HDI 2022

| Statistics         | HDI (2022) |
|--------------------|------------|
| Mean               | 0.7229     |
| Median             | 0.7395     |
| Standard Deviation | 0.1530     |
| Minimum            | 0.3800     |
| Maximum            | 0.9670     |

The mean HDI of 0.7229 falls between High and Very High development categories. The standard deviation of 0.1530 shows substantial global difference. The 0.5870 gap between highest and lowest HDI reflects significant inequality in human development worldwide.

## 1.3: Top and Bottom Performers

**Highest HDI Countries (2022):** Switzerland leads with approximately 0.96, followed by the United States (0.92). These countries demonstrate strong performance across all three HDI components, with particularly high average income and life span.

**Lowest HDI Countries (2022):** Nepal records the lowest HDI at approximately 0.5, reflecting ongoing challenges in health infrastructure and education access. This stark contrast with high-HDI nations highlights global development disparities.

## 1.4: HDI Categories & Distribution

After classifying countries using UNDP thresholds:

- **Very High ( $\geq 0.800$ ):** Switzerland, United States
- **High (0.700-0.799):** China
- **Medium (0.550-0.699):** India
- **Low ( $< 0.550$ ):** Nepal

This distribution proves that while many countries have attained high development, a significant portion still faces development challenges.

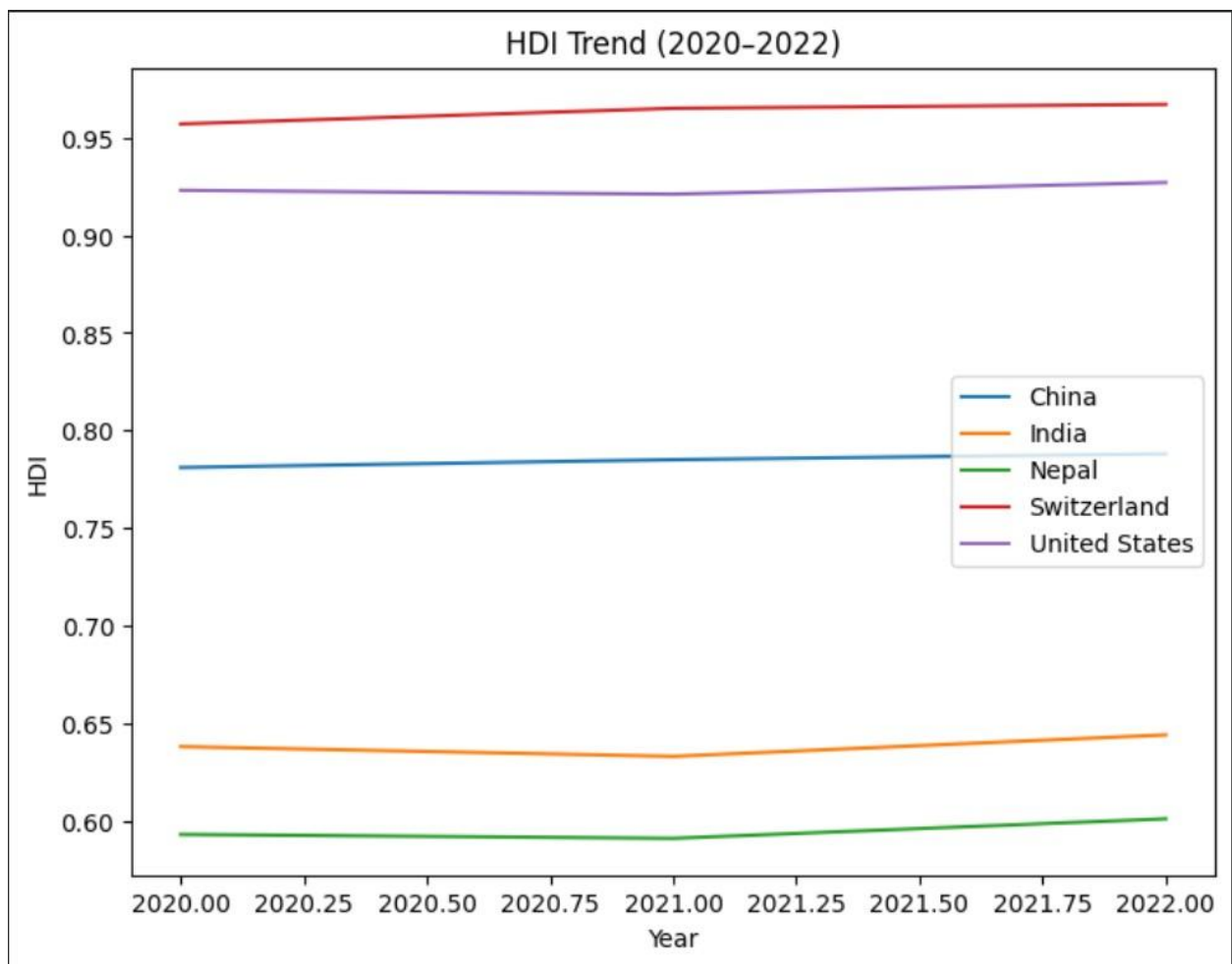
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## Problem 1B

### HDI Trend Analysis (2020-2022)

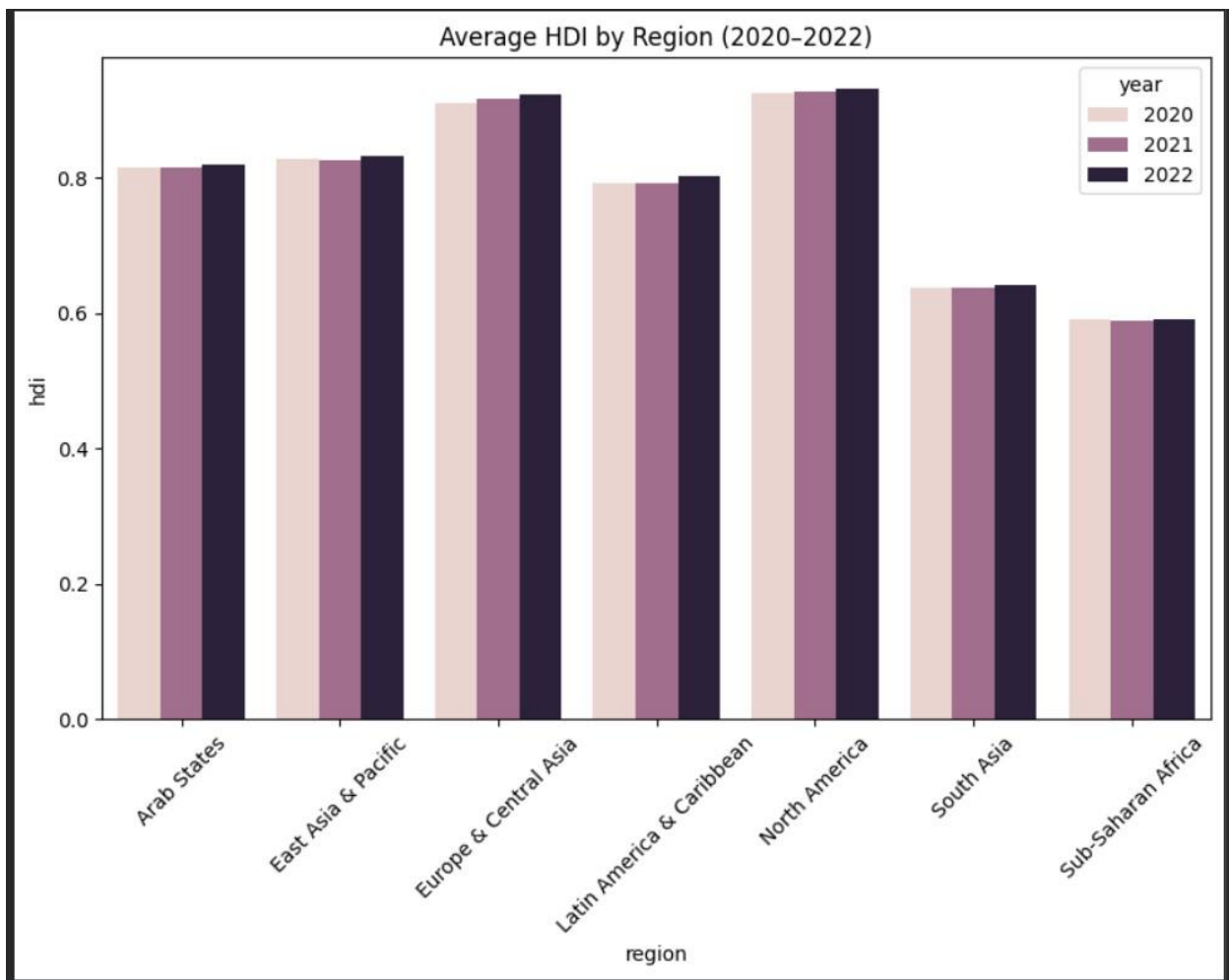
We extracted HDI data for 2020-2022, creating a three-year panel of 206 countries. This period captures the impact of the COVID-19 pandemic on global development. Missing values were handled through systematic examination of patterns.

Line Chart:



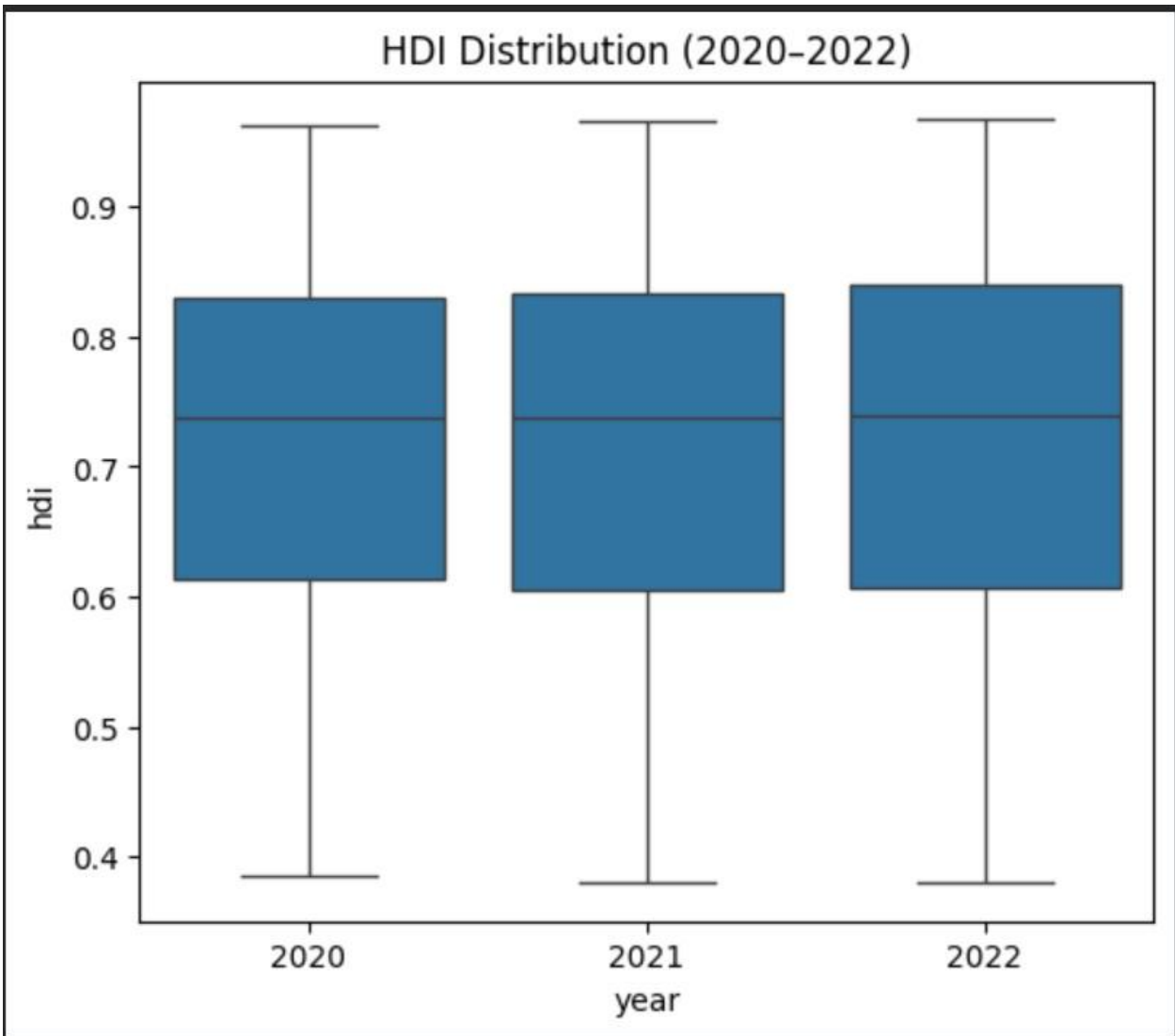
This figure displays HDI trends for China, India, Nepal, Switzerland, and the United States of America. You can see that China shows a consistent trajectory advancing from 2020 to 2022. Nepal shows a decreasing value from 2020 to 2021 but it shows a small improvement from 2021 to 2022. In contrast, Switzerland exhibits the exact opposite trend. From 2020 to 2021, it shows a drastic improvement in the trajectory but it slightly starts to decrease from 2021 to 2022.

#### Bar Chart:



This figure compares regional average HDI across the three-year period. North America and Europe & Central Asia maintain two of the highest average values, while Arab States, East Asia & Pacific and Latin America & Caribbean are neck to neck followed by South Asia and Sub-Saharan Africa.

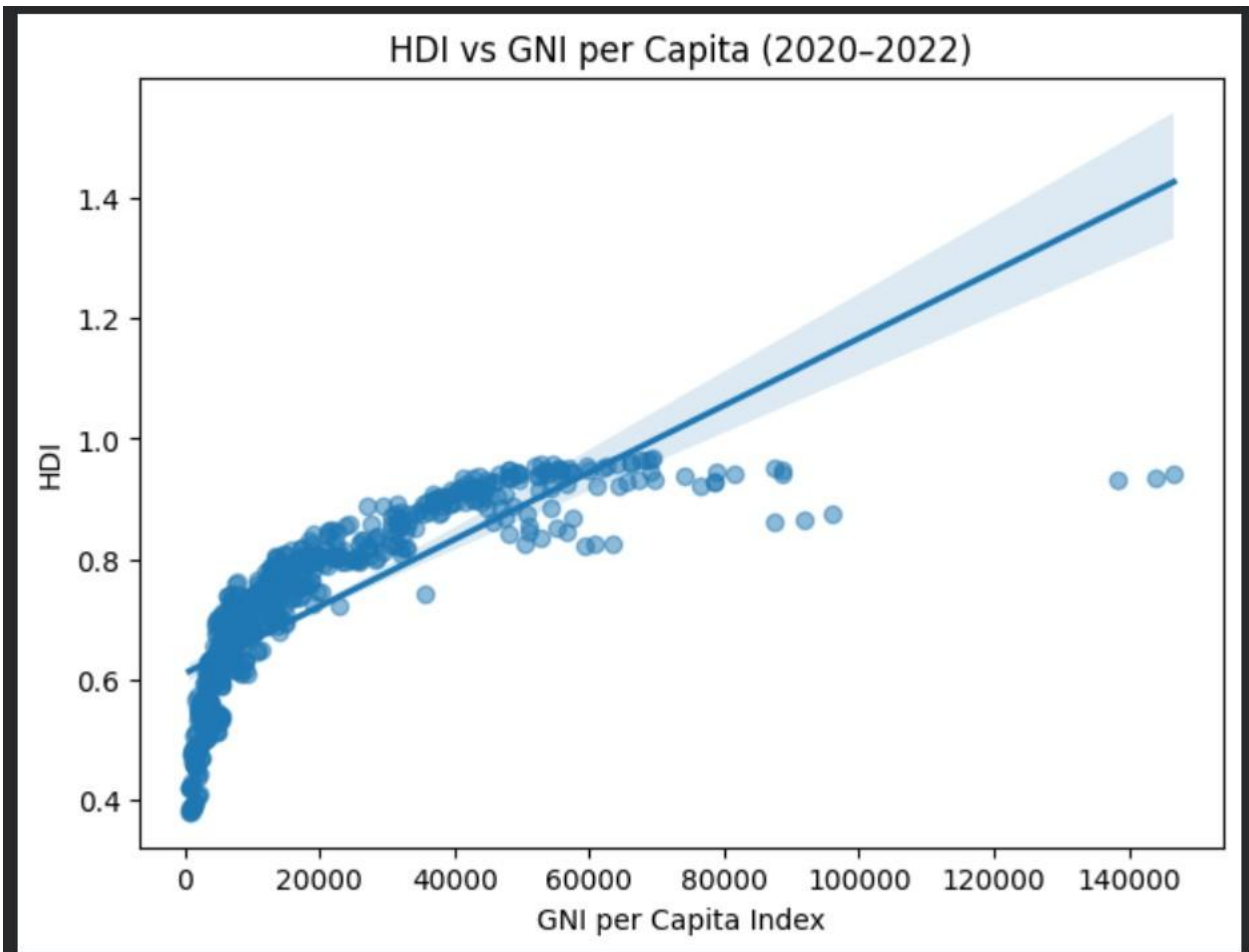
**Box plot:**



In this figure, the distribution reveals the spread of HDI across the three years. The interquartile range (IQR) in 2022 is approximately 0.7.



### Scatter Plot:

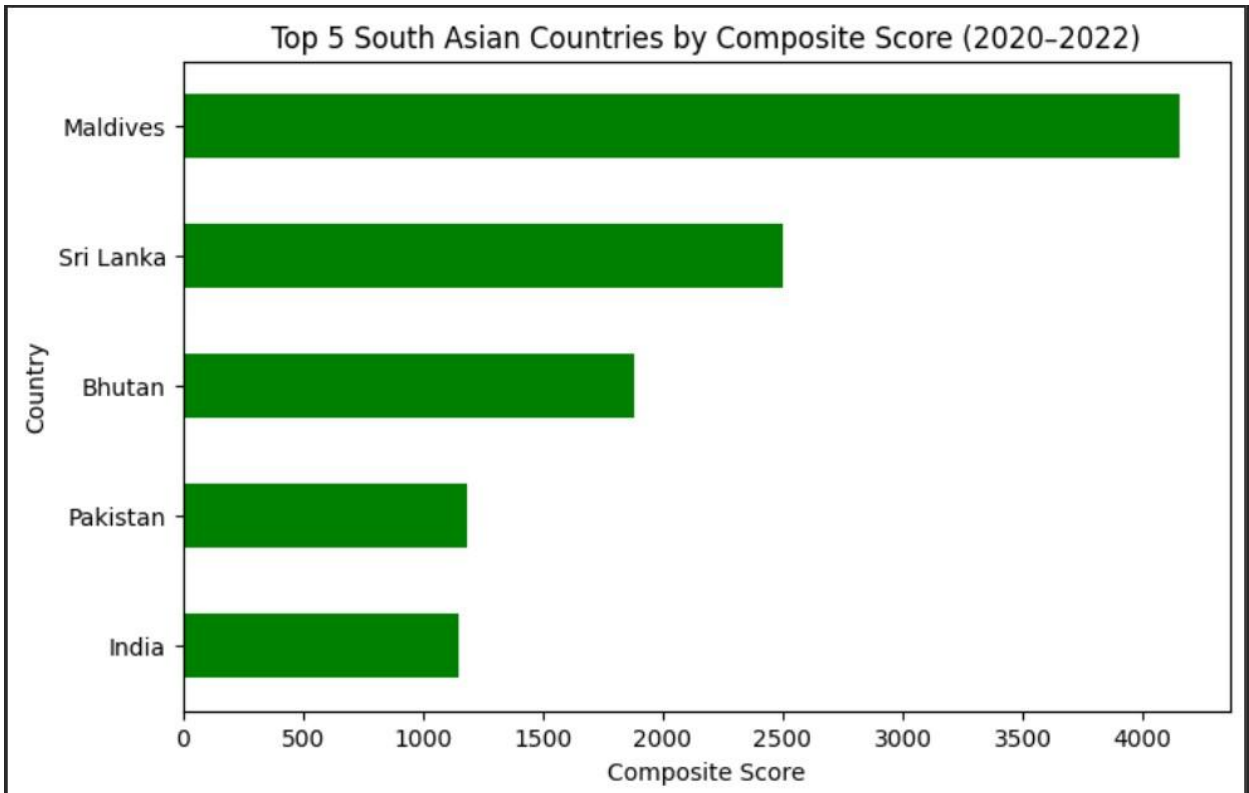


The scatter plot chart demonstrates an effective correlation between GNI per capita and HDI, suggesting economic prosperity contributes to human development. However, several countries deviate from this trend, indicating that income alone doesn't determine development outcomes.

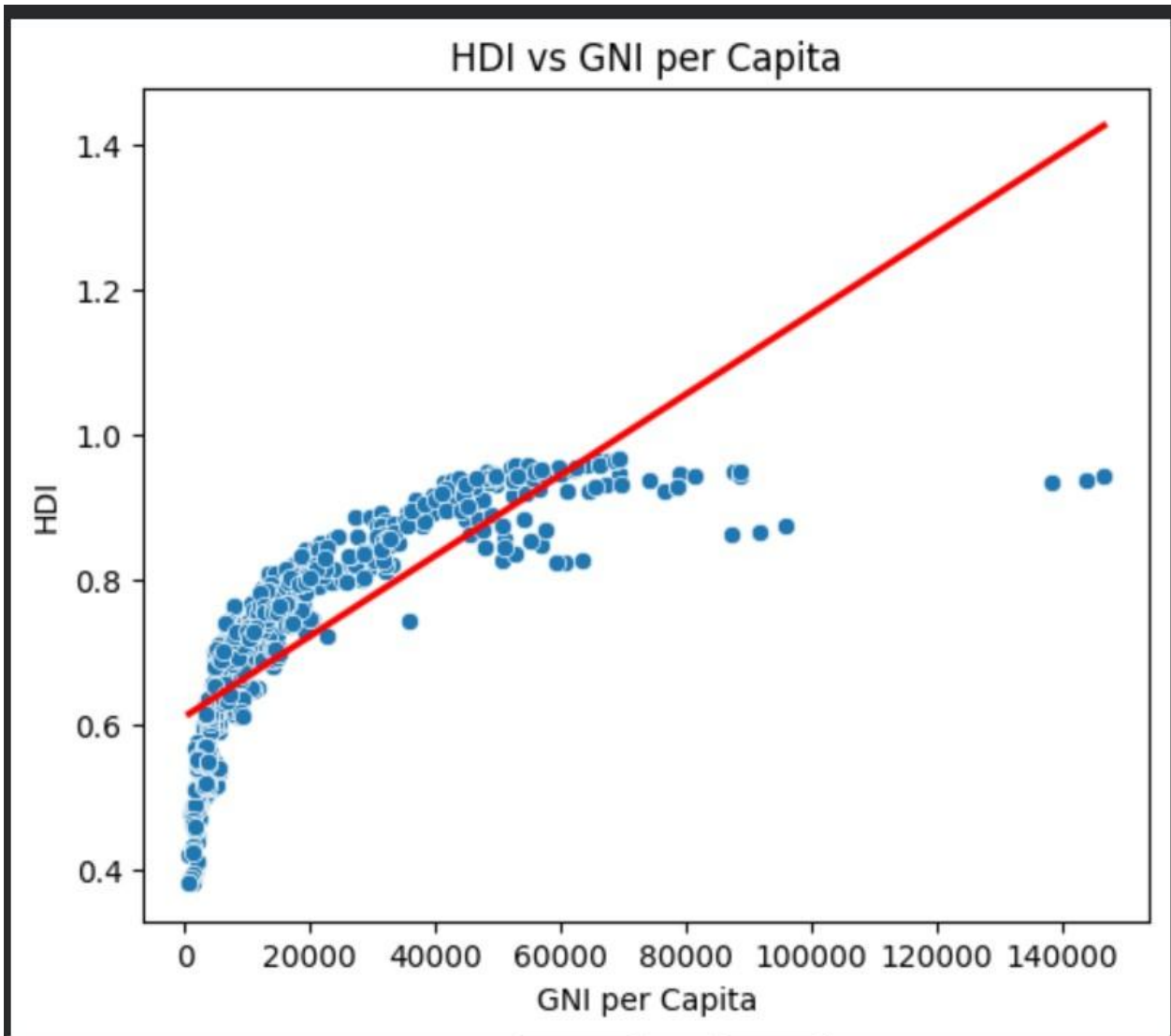
### Problem 2 - Advanced HDI Exploration

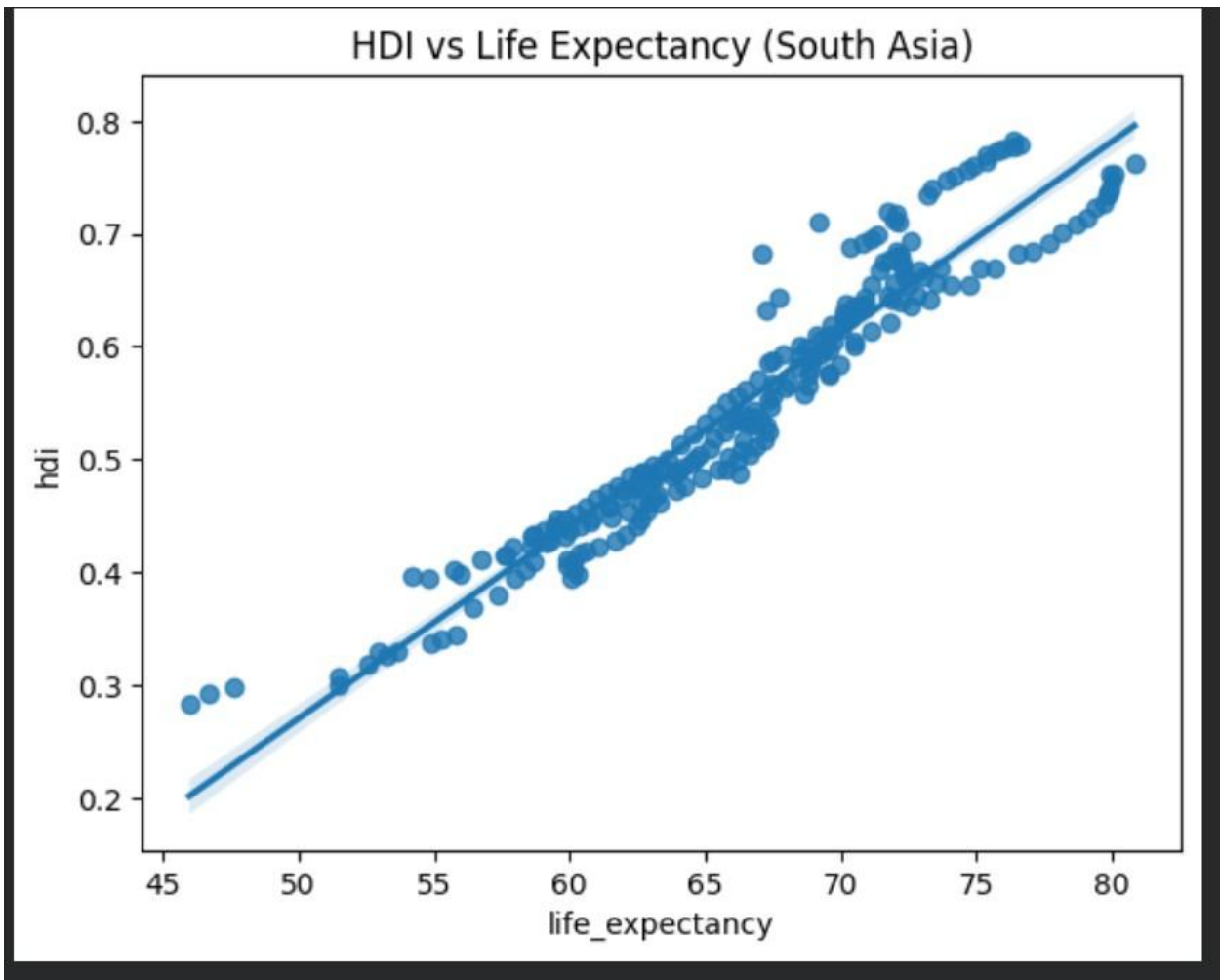
**Top 5 by Composite Score:** Maldives with a composite score of above 4000 followed by Sri Lanka (above 2500), Bhutan (slightly less than 2000), then Pakistan and India (approximately around 1000).

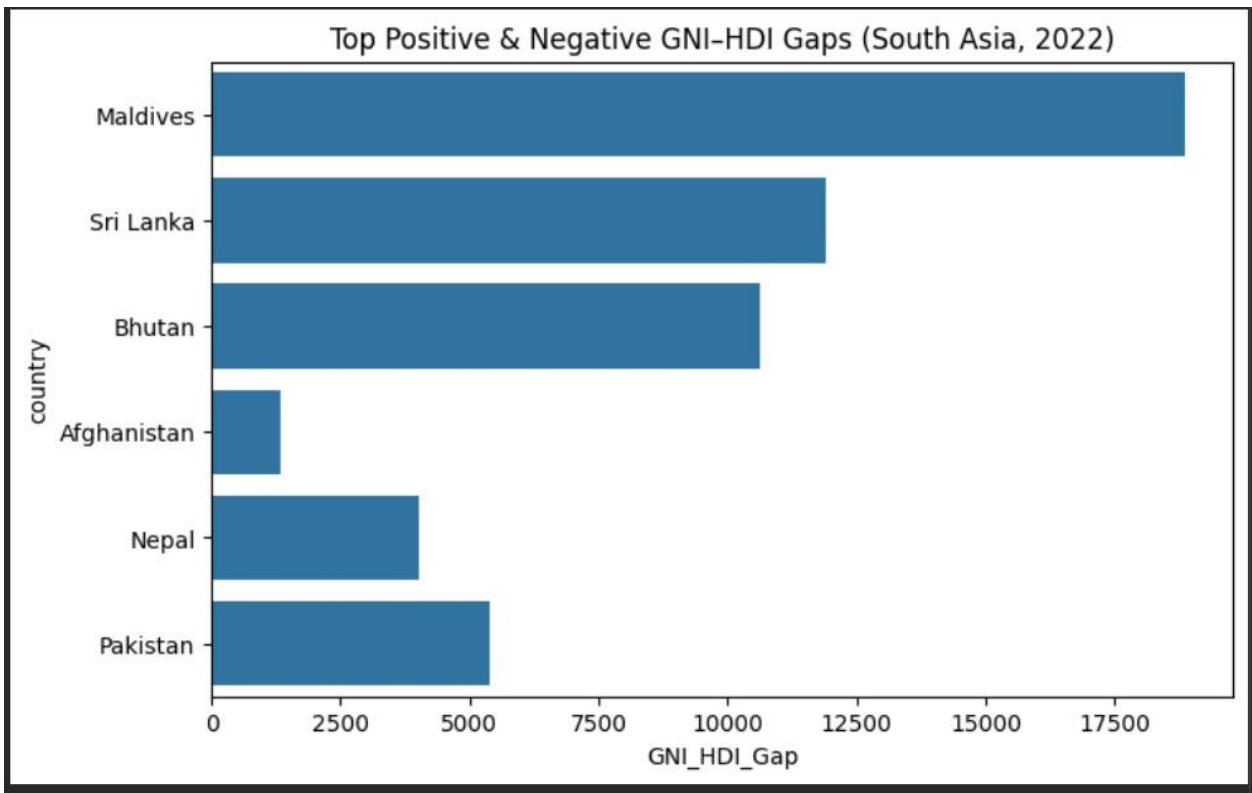
**Bar Chart:**



**Scatter Plot:**

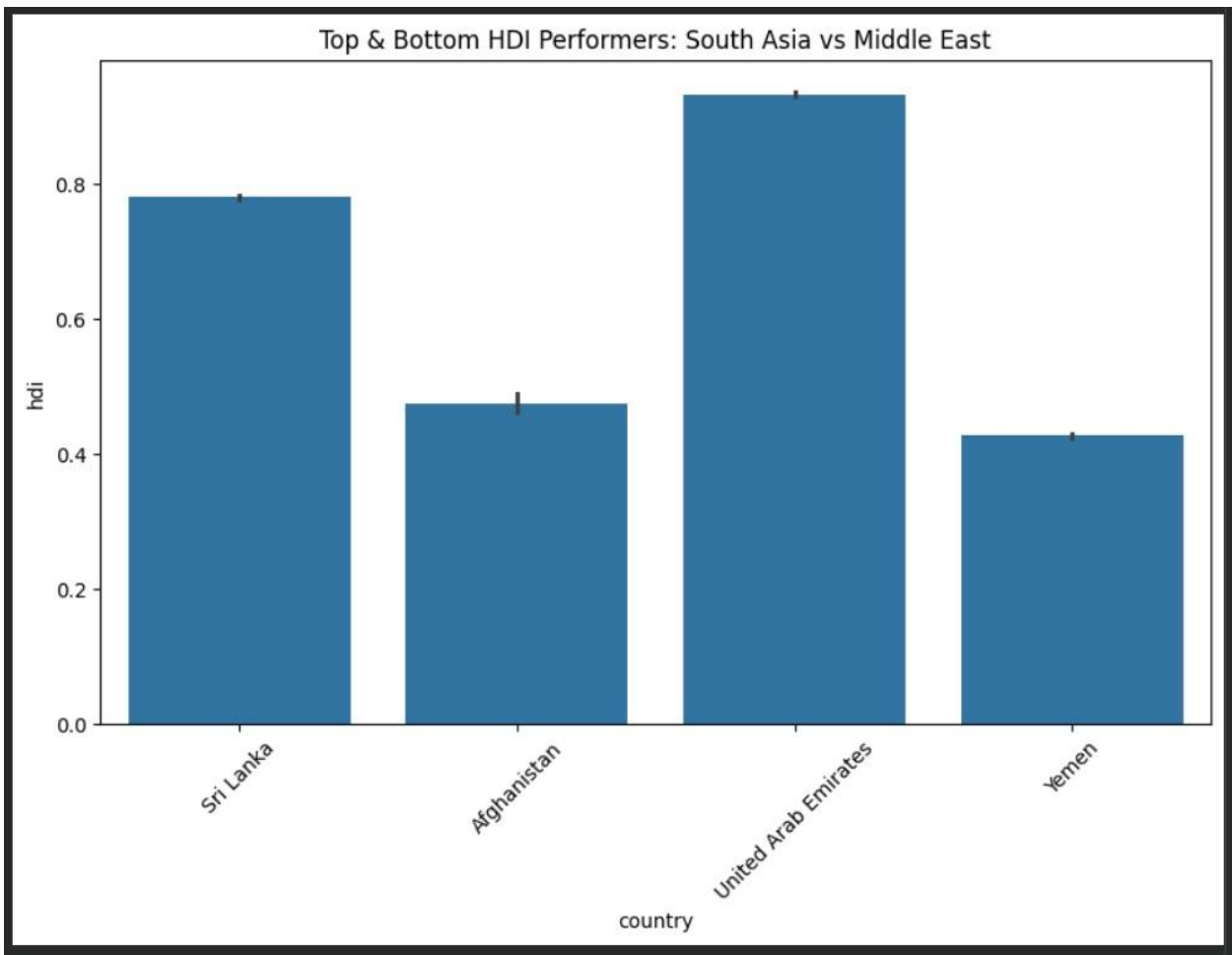






### 9 Problem 3

## Comparison of South Asia and Middle East

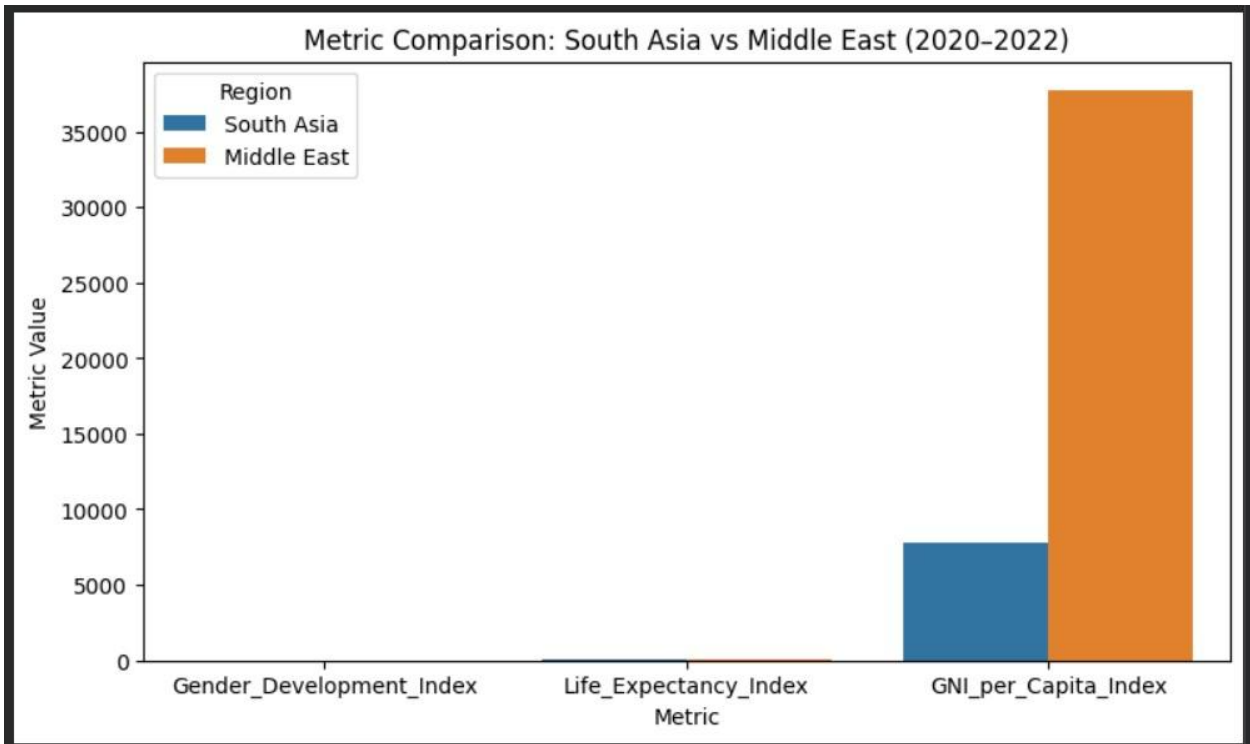


### Top and Bottom Performers

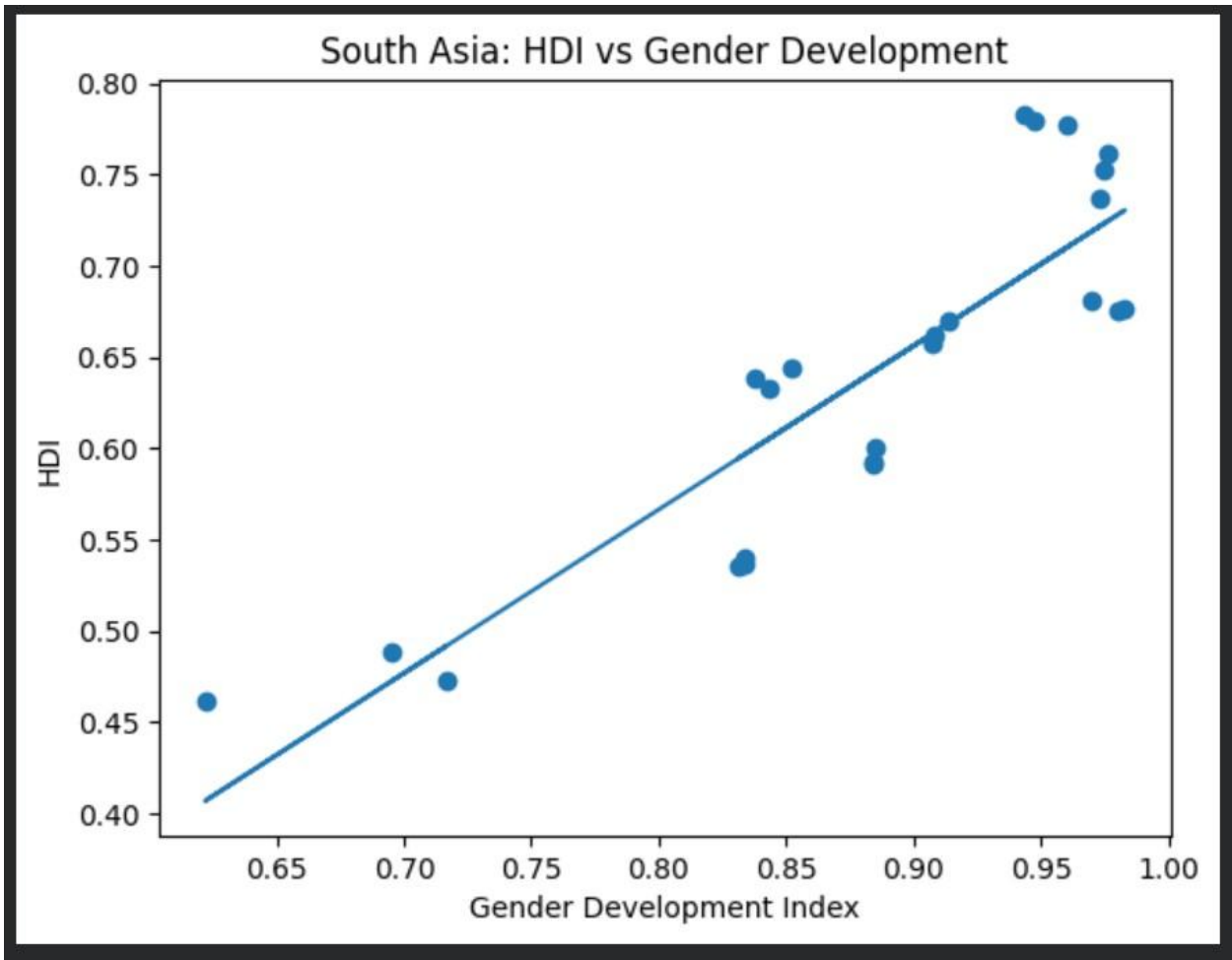
**Top 2:** United Arab Emirates (Middle East) and Sri Lanka (South Asia)

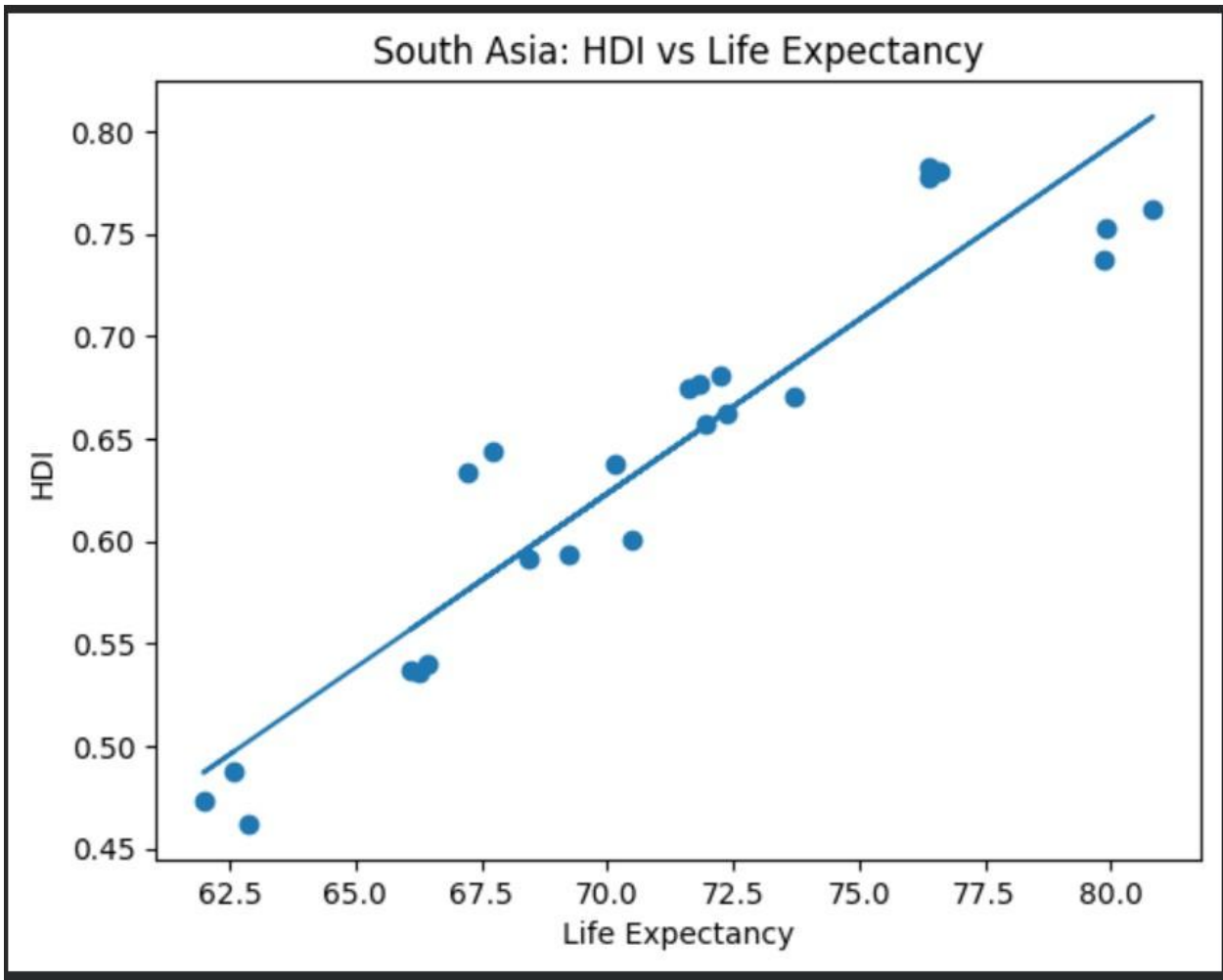
**Bottom 2:** Afghanistan (South Asia) and Yemen (Middle East)

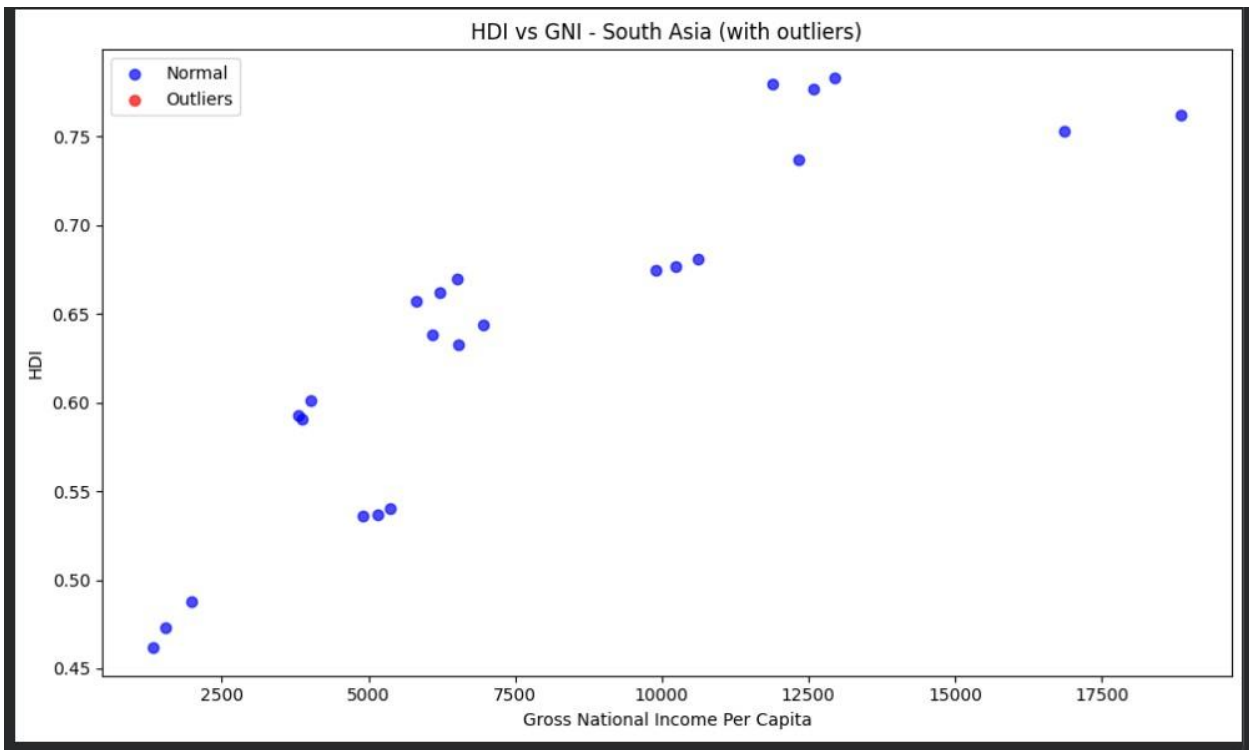
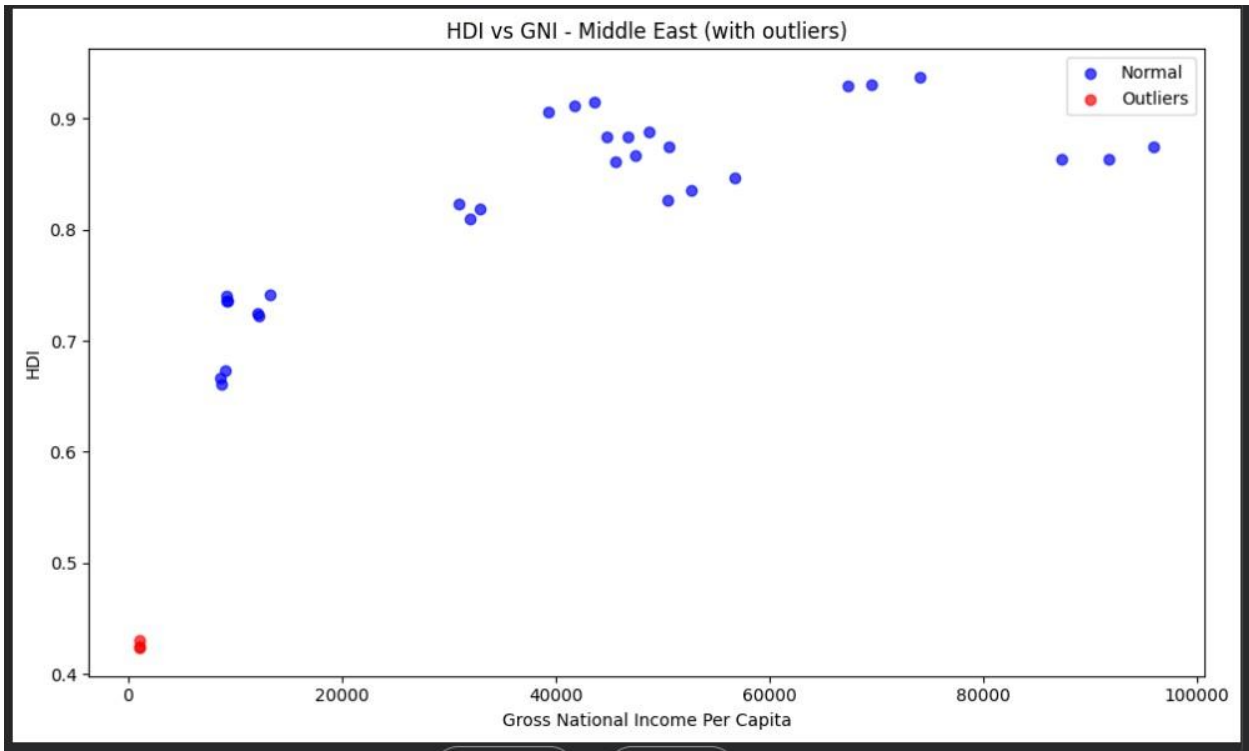
### Metric Comparisons:











Conclusion:

This study reveals significant global HDI disparities. The 2020-2022 period showed pandemic impacts with partial recovery. South Asia and the Middle East differ substantially in growth levels and internal variation. Life expectancy emerged as the strongest HDI predictor. Limitations include missing data for some countries and the simplified nature of composite indices. Future research should examine longer time periods and additional socioeconomic factors.

## <sup>1</sup>References:

United Nations Development Programme. (2023). Human Development Report UNDP. Lecture, Tutorial, Workshop.



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